★ determining ideal EA frameworks that fit business needs

WEEK 11 • IS321 ENTERPRISE ARCHITECTURE

The Most Significant EA Frameworks

The Zachman Framework in Depth — and How It Compares to the Others

TODAY'S ROADMAP

What We'll Cover

01	**Why So Many Frameworks?**	About 20 exist; most enterprises pick from a handful of well-established ones
02	**The Zachman Framework**	Origins, structure, the six perspectives, and the six questions
03	**Quick Recaps: TOGAF, EA³ Cube, DoDAF**	Enough context to compare them meaningfully — full depth comes in other weeks
04	**Framework Comparison**	Origin, structure, and best-fit use case, side by side
05	**Choosing the Right Framework**	Practical guidance for matching a framework to a real situation

federal frameworks operate on a B2G basis → not widely used in academia contexts

FOUNDATIONS

Why So Many Frameworks?

About 20 EA frameworks exist. The most well-known:

The Zachman Framework

A classification schema — covered in depth this week

TOGAF

An architecture development method — Week 12

EA³ Cube (Bernard)

This course's primary framework — Weeks 3, 5, 14

DoDAF

The U.S. Department of Defense's framework

FEA

The Federal Enterprise Architecture framework

Gartner & Hybrid Approaches

Vendor and organization-specific adaptations

IN DEPTH

The Zachman Framework — Origins

↓ Master file → library documenting complex architecture → what to document NOT how and classifies them

> *The Zachman Framework™ is a schema — the intersection between two historical classifications that have been in use for literally thousands of years.*

The First Classification

The fundamentals of communication found in the primitive interrogatives: What, How, When, Who, Where, and Why. Integrating answers to these questions enables a comprehensive, composite description of complex ideas.

The Second Classification

Derived from reification — the transformation of an abstract idea into a concrete instantiation, a concept first postulated by ancient Greek philosophers.

IN DEPTH

Six Perspectives, Six Questions

A two-dimensional matrix: six rows (perspectives) and six columns (fundamental questions). The intersecting cells describe the enterprise in a detailed, structured way — and every row and column must stay in place to keep the holistic overview intact.

Rows: The Perspectives (Who's Asking)

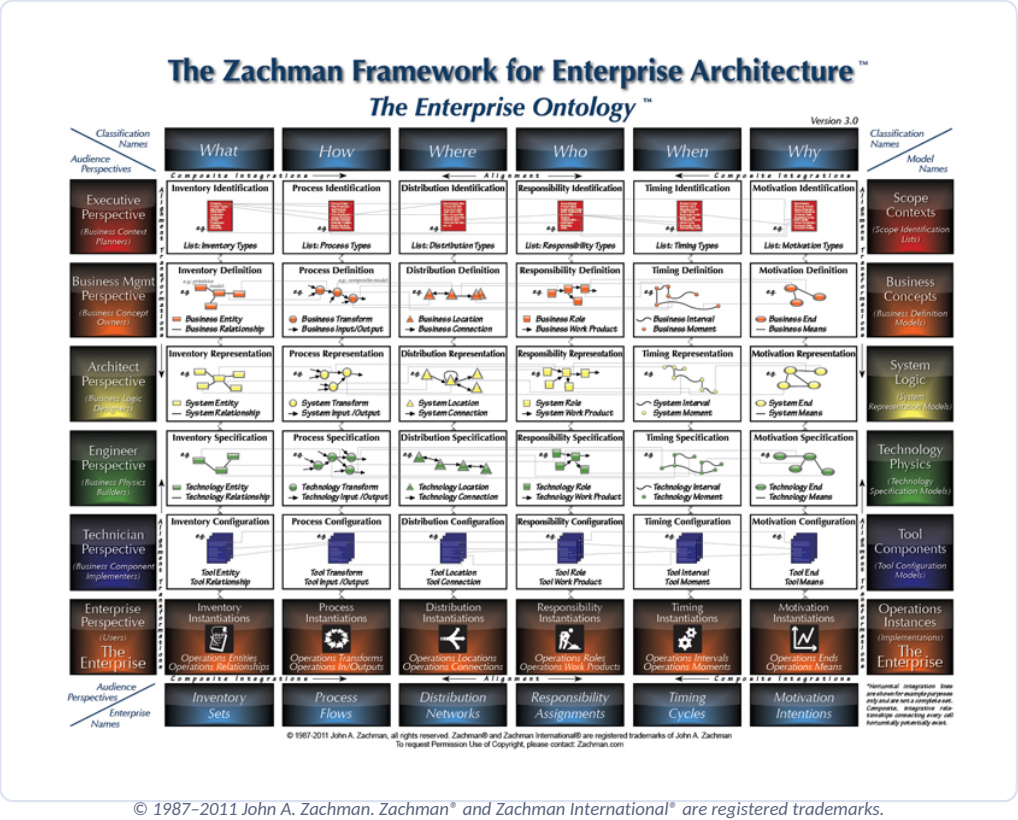
- Executive — size and cost
- Business Management — process interaction
- Architect — how software represents the model
- Engineer — applying technology to problems
- Technician — following instructions
- Enterprise — the operational system itself

Columns: The Questions (What's Asked)

- What? — the enterprise's data
- How? — functions and processes
- Where? — locations and logistics
- Who? — organizational units and people
- When? — business cycle and events
- Why? — goals and strategy translated to means

IN DEPTH

The Official Zachman Framework Grid



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IN DEPTH

When Would You Use Zachman?

↓ solely serves as a documentation-only tool

↓ easily combined with other frameworks

Focus Without Losing the Whole

Concentrate on independent objects without losing sight of the holistic perspective and relationships between them.

↓ reduces ambiguity and the need to clarify

Stakeholder Conversations

Clarify what each level should focus on with stakeholders, without delving into technical detail.

Documentation Tool

Combines easily with other models — it's a classification tool, not a competing methodology.

Simplifying Team Communication

Gives the whole team a shared structure for describing what they're working on.

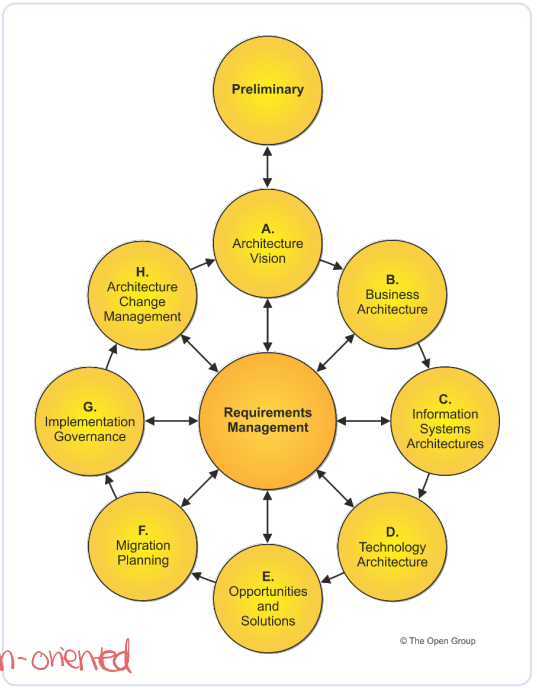
Gap Analysis

Look at what perspectives you've already described, and see what's still missing.

QUICK RECAP — FULL DEPTH IN WEEK 12

TOGAF

The Open Group Architecture Framework. Developed by 300+ enterprise architects from companies like Dell, Cognizant, and Microsoft, dating to 1995. Its Architecture Development Method (ADM) is a step-by-step, cyclical process — with Requirements Management at the center, reflecting the ongoing alignment of every phase.



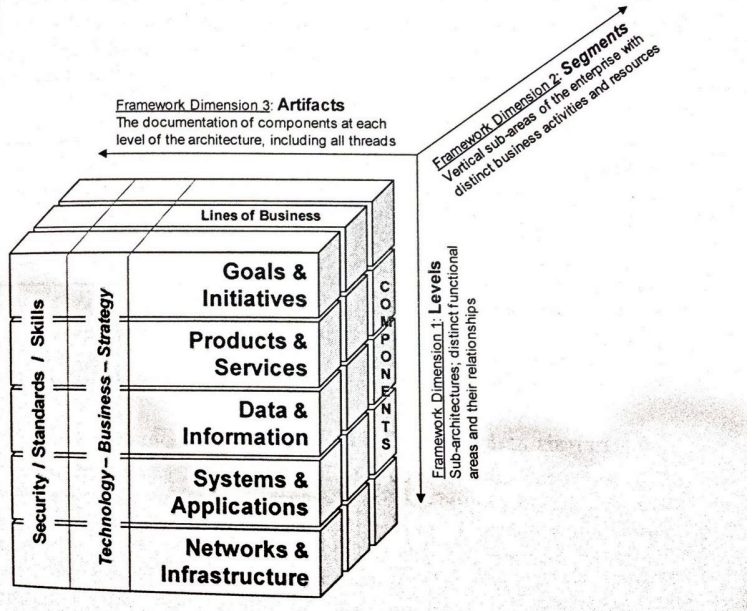
Adopted by 60% of Fortune 500 companies

Highly adaptable and collaborative, but criticized for being theory-focused. Use it as a guide, not an actionable plan on its own.

QUICK RECAP — ALREADY COVERED IN WEEKS 3 & 5

EA³ Cube (Bernard)

This course's primary framework, developed by Scott Bernard. A three-dimensional model: five hierarchical levels (Goals & Initiatives through Networks & Infrastructure), Segments as a second dimension (Lines of Business), and Artifacts as a third — with Security/Standards/Skills running through every level as a thread.



Already familiar to you

You met this in Week 3 (the five levels), used it again in Week 5 (the Repository), and will revisit it in Week 13, with AI-tooling depth explored further in Week 14.

QUICK RECAP

DoDAF



The U.S. Department of Defense Architecture Framework — provides visualization infrastructure for specific stakeholder concerns through viewpoints organized by various views. These views are artifacts for understanding a broad, complex architecture through tabular, structural, behavioral, pictorial, or graphical means.

Three core views:



Operational View

How the mission actually operates

System / Service View

The systems and services that support it

Technical Standard View

The technical rules everything must follow

THE COMPARISON

Four Frameworks, Side by Side

	Origin	Structure	Best Fit
Zachman	John Zachman, 1987 — a classification schema, not a method	6×6 matrix: perspectives × fundamental questions	Comprehensive documentation & stakeholder communication
TOGAF	The Open Group, 1995 — an architecture development method	9-phase cyclical ADM, with Requirements Management at the center	Large-enterprise transformation with a defined, repeatable process
EA³ Cube	Scott Bernard, 2005 — an academic/practitioner framework	5 levels × Segments × Artifacts, with Threads running through	Organizations wanting a structured repository and management plan (this course's approach)
DoDAF	U.S. Dept. of Defense — a viewpoints-and-views framework	Operational, System/Service, and Technical Standard views	Defense and large government systems with strict interoperability needs

PRACTICAL GUIDANCE

Choosing the Right Framework

"I need to make sure I haven't missed documenting something."	**Zachman — its completeness guarantee (all 36 cells) is exactly built for this.**
"I need a repeatable process to actually build the architecture."	**TOGAF — the ADM gives you an explicit sequence of phases to follow.**
"I want a structured repository and an ongoing management plan."	**EA³ Cube — this is the approach this entire course has built around.**
"I'm working in defense or a large government system with strict interoperability rules."	**DoDAF — purpose-built for exactly this context.**

CHAPTER SUMMARY

Key Takeaways

- About 20 EA frameworks exist; most enterprises work from a small handful of well-established ones.
- The Zachman Framework is a classification schema (a 6×6 matrix of perspectives × questions), not a step-by-step method — it organizes what to document, not how to build it.
- TOGAF's ADM, EA³ Cube, and DoDAF each answer a different question: how to build it, how to manage/repository it, and how to handle strict government interoperability, respectively.
- Zachman and TOGAF are complementary, not competing — many real implementations use Zachman's completeness check alongside TOGAF's process.
- Choosing a framework is a fit question, not a best/worst question — match the framework to what you actually need to accomplish.

COMING UP NEXT

Further Reading & Next Session

Reading Material

- Zachman International — zachman.com/about-the-zachman-framework
- The Open Group — TOGAF Standard, Version 9.2
- Bernard, S. (2012). An Introduction to Enterprise Architecture (3rd ed.)

Next: Week 12

The Open Group's TOGAF and the Architecture Development Method (ADM) — in full depth this time. Mapped to CLO1, CLO4.

Questions?